

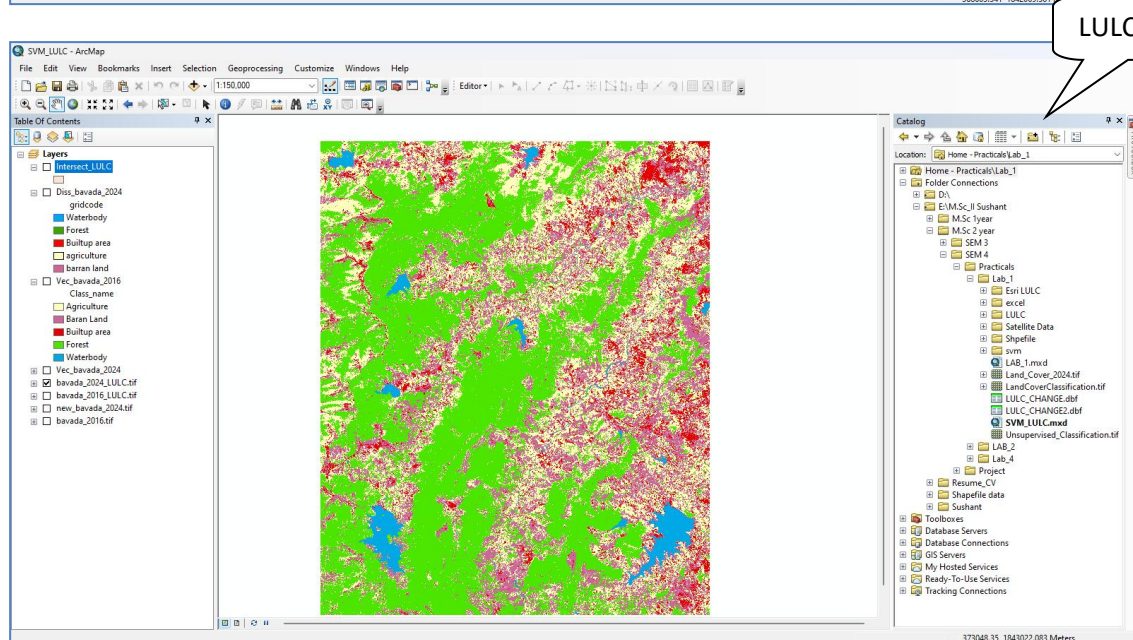
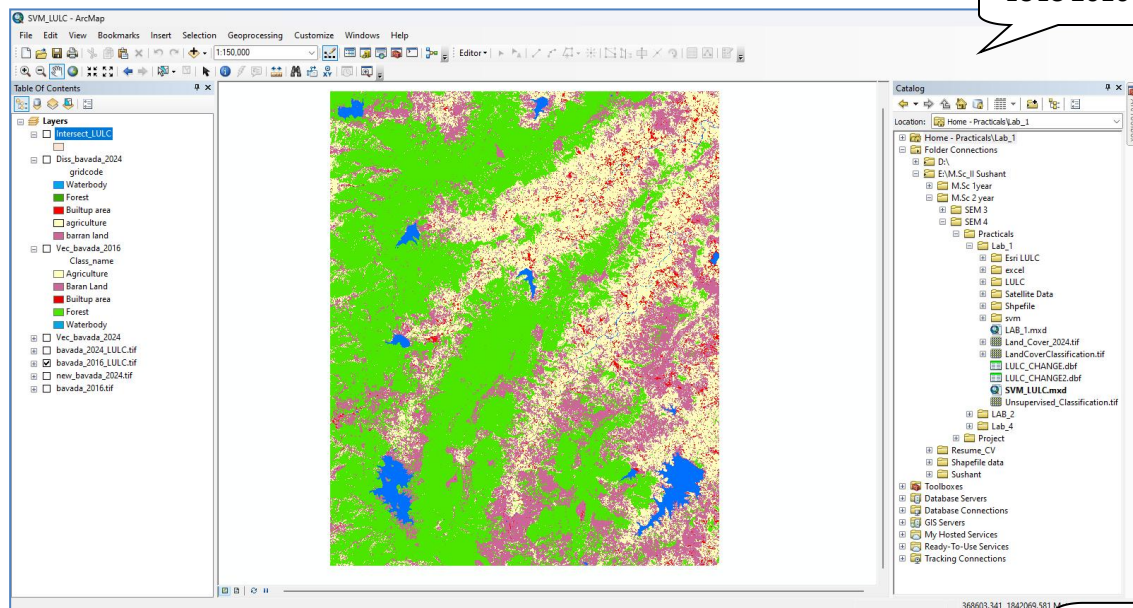
Lab 1

Change Detection Analysis

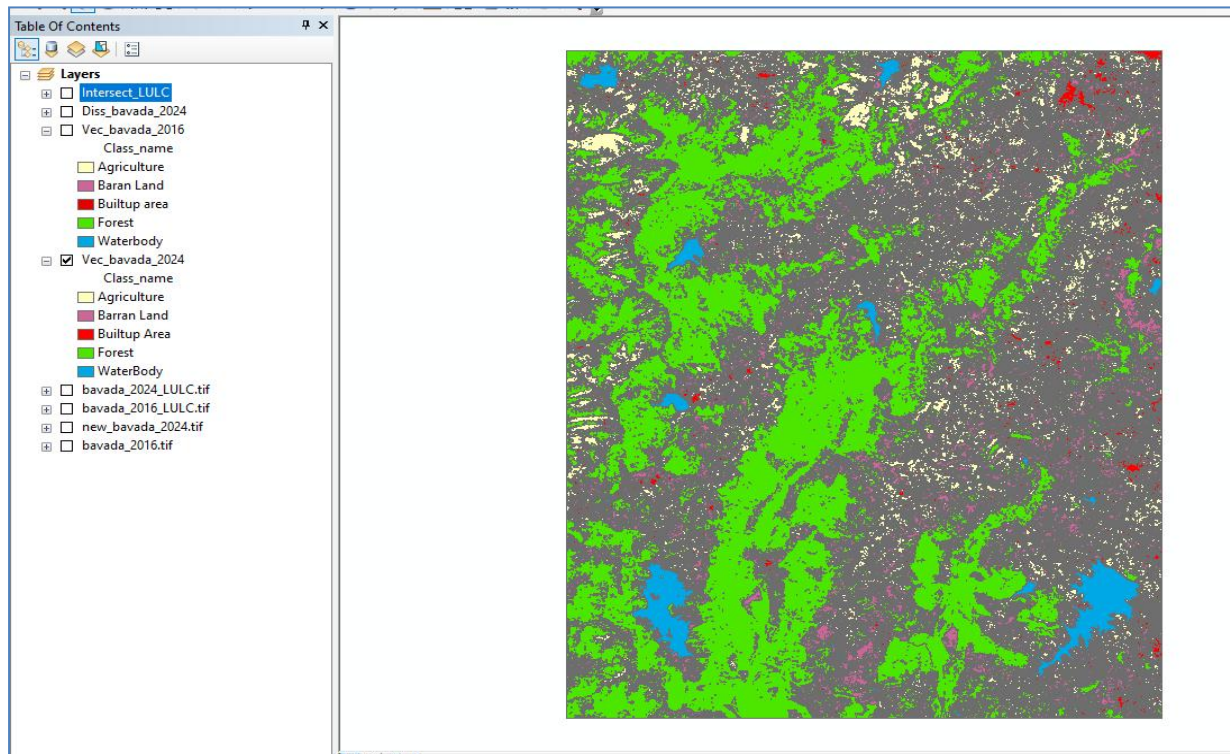
Student Name : Sushant Bhikaji Gurav

Change detection analysis is a method for identifying, describing, and quantifying changes in images of the same area over time. For this analysis we choose the Land Use Land cover for change detection

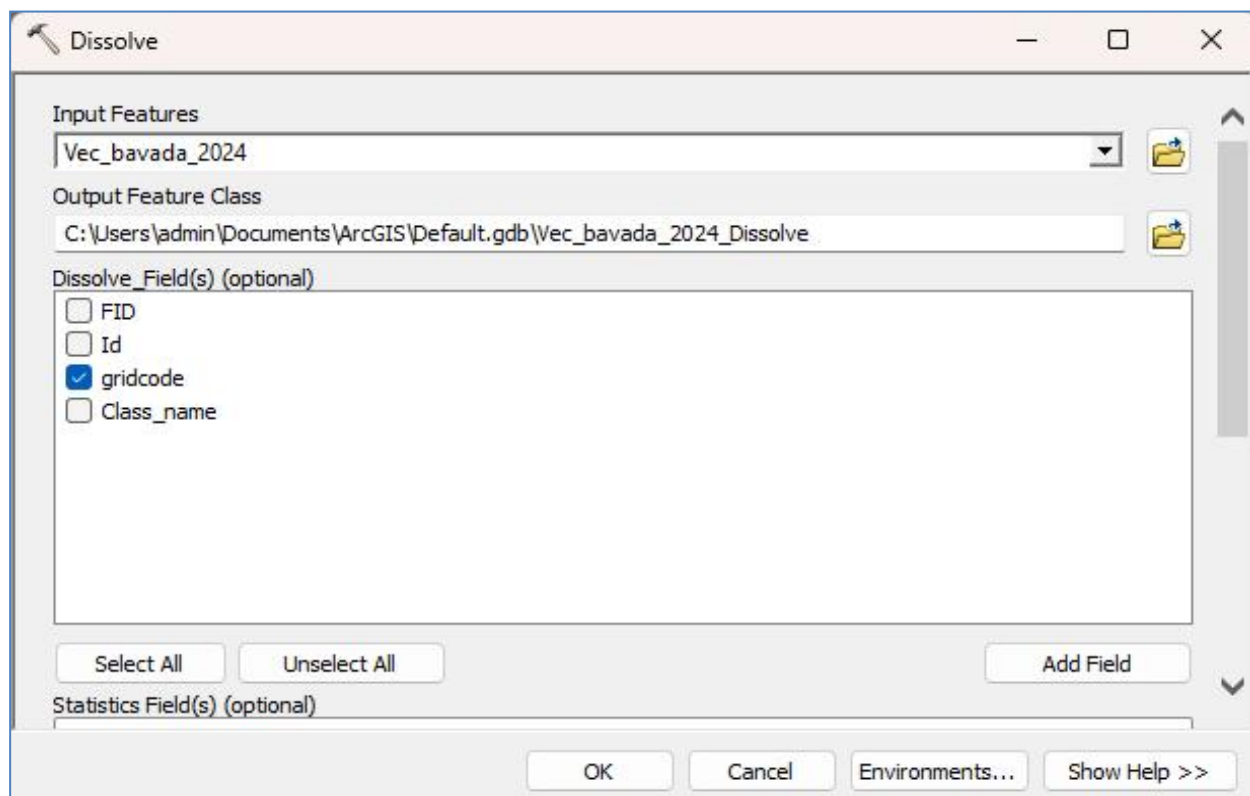
Step 1) Make LULC of 2016 and 2024



Step 2) Convert Both LULC into Vector



Step 3) Dissolve All classes



Dissolve Classes

Table					
Diss_bavada_2024					
	FID	Shape *	gridcode	LULC_2024	Area_2024
▶	0	Polygon	1	waterbody	14.7741
	1	Polygon	2	forest	255.288
	2	Polygon	3	builtup area	33.4382
	3	Polygon	4	agriculture	155.14
	4	Polygon	5	barran land	123.662

Table



Vec_bavada_2016

	FID	Shape *	Id	gridcode	LULC_2016	Area_2016
▶	3	Polygon	26	3	Builtup area	11.0207
	4	Polygon	111	1	Waterbody	14.6115
	0	Polygon	1	5	Baran Land	133.836
	2	Polygon	4	4	Agriculture	151.814
	1	Polygon	3	2	Forest	271.02

Step 4) Intersect the both LULC Layer

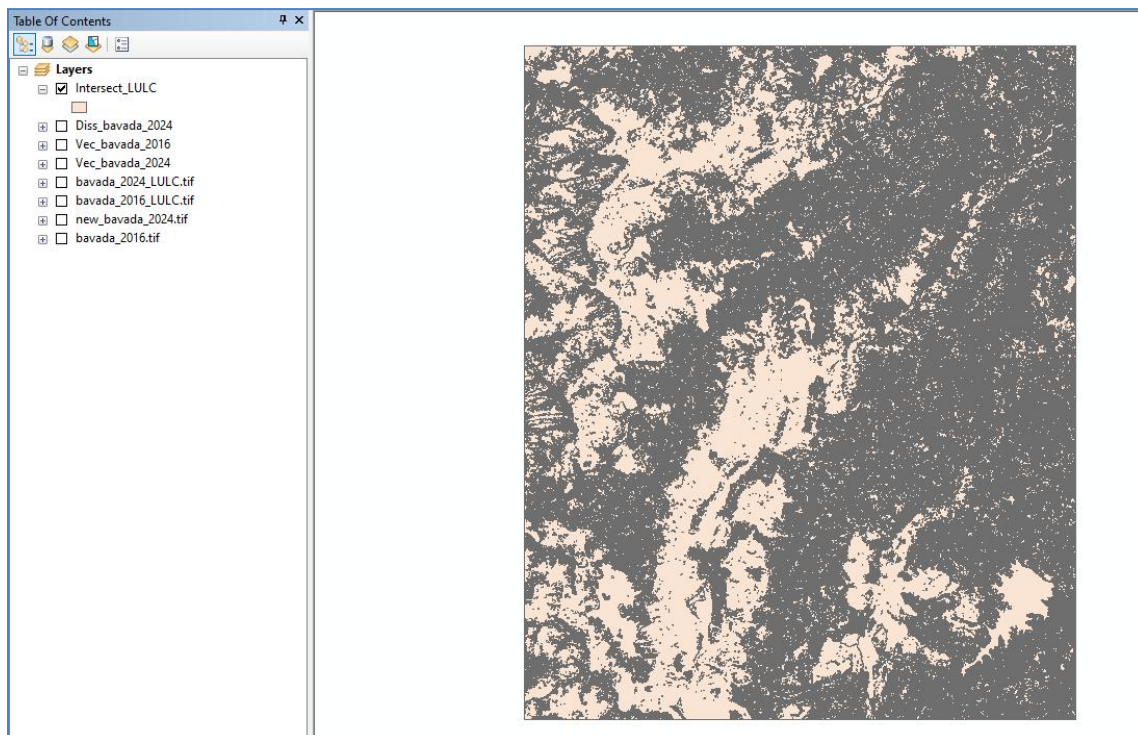


Table											
Intersect_LULC											
	FID	Shape *	FID_Diss_b	gridcode	LULC_2024	Area_2024	FID_Vec_ba	Id	gridcode_1	LULC_2016	Area_2016
▶	0	Polygon	0	1	waterbody	14.7741	0	1	5	Baran Land	133.836
	1	Polygon	0	1	waterbody	14.7741	1	3	2	Forest	271.02
	2	Polygon	0	1	waterbody	14.7741	2	4	4	Agriculture	151.814
	3	Polygon	0	1	waterbody	14.7741	3	26	3	Builtup area	11.0207
	4	Polygon	0	1	waterbody	14.7741	4	111	1	Waterbody	14.6115
	5	Polygon	1	2	forest	255.288	0	1	5	Baran Land	133.836
	6	Polygon	1	2	forest	255.288	2	4	4	Agriculture	151.814
	7	Polygon	1	2	forest	255.288	3	26	3	Builtup area	11.0207
	8	Polygon	1	2	forest	255.288	4	111	1	Waterbody	14.6115
	9	Polygon	2	3	builtup area	33.4382	0	1	5	Baran Land	133.836
	10	Polygon	2	3	builtup area	33.4382	1	3	2	Forest	271.02
	11	Polygon	2	3	builtup area	33.4382	2	4	4	Agriculture	151.814
	12	Polygon	2	3	builtup area	33.4382	4	111	1	Waterbody	14.6115
	13	Polygon	3	4	agriculture	155.14	0	1	5	Baran Land	133.836
	14	Polygon	3	4	agriculture	155.14	1	3	2	Forest	271.02
	15	Polygon	3	4	agriculture	155.14	3	26	3	Builtup area	11.0207
	16	Polygon	3	4	agriculture	155.14	4	111	1	Waterbody	14.6115
	17	Polygon	4	5	barran land	123.662	1	3	2	Forest	271.02
	18	Polygon	4	5	barran land	123.662	2	4	4	Agriculture	151.814
	19	Polygon	4	5	barran land	123.662	3	26	3	Builtup area	11.0207
	20	Polygon	4	5	barran land	123.662	4	111	1	Waterbody	14.6115

Step 5) Add field with name LULC CHANGE & Area Change

×

Add Field

Name:

LULC_CHANGE

Type:

Text

Field Properties

Length

50

OK

Cancel

×

Add Field

Name:

Area_Change

Type:

Float

Field Properties

Precision

0

Scale

0

OK

Cancel

Step 6) Right click on LULC_Change Field and click on field calculator and apply formula

[LULC_2016] + "-" + [LULC_2024]

Field Calculator

Parser
☒ VB Script ☐ Python

Fields:
 LULC_2024
 Area_2024
 FID_Vec_ba
 Id
 gridcode_1
 LULC_2016
 Area_2016
 LULC_chang
 Area_chang

Type:
☒ Number
☐ String
☐ Date

Functions:
 Abs ()
 Atn ()
 Cos ()
 Exp ()
 Fix ()
 Int ()
 Log ()
 Sin ()
 Sqr ()
 Tan ()

☐ Show Codeblock

LULC_chang =
 [LULC_2016] + "-" + [LULC_2024]

[About calculating fields](#)

Clear Load... Save...

OK Cancel

Table											
Intersect_LULC											
FID	Shape *	FID_Diss_b	gridcode	LULC_2024	Area_2024	FID_Vec_ba	Id	gridcode_1	LULC_2016	Area_2016	LULC_chang
0	Polygon	0	1	waterbody	14.7741	0	1	5	Baran Land	133.836	Baran Land -waterbody
1	Polygon	0	1	waterbody	14.7741	1	3	2	Forest	271.02	Forest -waterbody
2	Polygon	0	1	waterbody	14.7741	2	4	4	Agriculture	151.814	Agriculture -waterbody
3	Polygon	0	1	waterbody	14.7741	3	26	3	Builtup area	11.0207	Builtup area -waterbody
4	Polygon	0	1	waterbody	14.7741	4	111	1	Waterbody	14.6115	NO CHANGE
5	Polygon	1	2	forest	255.288	0	1	5	Baran Land	133.836	Baran Land -forest
6	Polygon	1	2	forest	255.288	2	4	4	Agriculture	151.814	Agriculture -forest
7	Polygon	1	2	forest	255.288	3	26	3	Builtup area	11.0207	Builtup area -forest
8	Polygon	1	2	forest	255.288	4	111	1	Waterbody	14.6115	Waterbody -forest
9	Polygon	2	3	builtup area	33.4382	0	1	5	Baran Land	133.836	Baran Land -builtup area
10	Polygon	2	3	builtup area	33.4382	1	3	2	Forest	271.02	Forest -builtup area
11	Polygon	2	3	builtup area	33.4382	2	4	4	Agriculture	151.814	Agriculture -builtup area
12	Polygon	2	3	builtup area	33.4382	4	111	1	Waterbody	14.6115	Waterbody -builtup area
13	Polygon	3	4	agriculture	155.14	0	1	5	Baran Land	133.836	Baran Land -agriculture
14	Polygon	3	4	agriculture	155.14	1	3	2	Forest	271.02	Forest -agriculture
15	Polygon	3	4	agriculture	155.14	3	26	3	Builtup area	11.0207	Builtup area -agriculture
16	Polygon	3	4	agriculture	155.14	4	111	1	Waterbody	14.6115	Waterbody -agriculture
17	Polygon	4	5	barran land	123.662	1	3	2	Forest	271.02	Forest -barran land
18	Polygon	4	5	barran land	123.662	2	4	4	Agriculture	151.814	Agriculture -barran land
19	Polygon	4	5	barran land	123.662	3	26	3	Builtup area	11.0207	Builtup area -barran land
20	Polygon	4	5	barran land	123.662	4	111	1	Waterbody	14.6115	Waterbody -barran land

Step 7) Right Click on Area Change and Click on Calculate Geometry to calculate Change area

Calculate Geometry

Property: Area

Coordinate System

☒ Use coordinate system of the data source:
PCS: WGS 1984 UTM Zone 43N

☐ Use coordinate system of the data frame:
PCS: WGS 1984 UTM Zone 43N

Units: Square Kilometers [sq km]

☐ Calculate selected records only
[About calculating geometry](#)

OK

Cancel

FID	Shape *	FID_Diss_b	gridcode	LULC_2024	Area_2024	FID_Vec_ba	Id	gridcode_1	LULC_2016	Area_2016	LULC_chang	Area_chang
0	Polygon	0	1	waterbody	14.7741	0	1	5	Baran Land	133.836	Baran Land -waterbody	0.417668
1	Polygon	0	1	waterbody	14.7741	1	3	2	Forest	271.02	Forest -waterbody	0.124203
2	Polygon	0	1	waterbody	14.7741	2	4	4	Agriculture	151.814	Agriculture -waterbody	0.104362
3	Polygon	0	1	waterbody	14.7741	3	26	3	Builtup area	11.0207	Builtup area -waterbody	0.084171
4	Polygon	0	1	waterbody	14.7741	4	111	1	Waterbody	14.6115	NO CHANGE	14.0437
5	Polygon	1	2	forest	255.288	0	1	5	Baran Land	133.836	Baran Land -forest	8.40852
6	Polygon	1	2	forest	255.288	2	4	4	Agriculture	151.814	Agriculture -forest	22.9163
7	Polygon	1	2	forest	255.288	3	26	3	Builtup area	11.0207	Builtup area -forest	0.103244
8	Polygon	1	2	forest	255.288	4	111	1	Waterbody	14.6115	Waterbody -forest	0.014104
9	Polygon	2	3	builtup area	33.4382	0	1	5	Baran Land	133.836	Baran Land -builtup area	17.2749
10	Polygon	2	3	builtup area	33.4382	1	3	2	Forest	271.02	Forest -builtup area	0.680293
11	Polygon	2	3	builtup area	33.4382	2	4	4	Agriculture	151.814	Agriculture -builtup area	10.9868
12	Polygon	2	3	builtup area	33.4382	4	111	1	Waterbody	14.6115	Waterbody -builtup area	0.029732
13	Polygon	3	4	agriculture	155.14	0	1	5	Baran Land	133.836	Baran Land -agriculture	36.1172
14	Polygon	3	4	agriculture	155.14	1	3	2	Forest	271.02	Forest -agriculture	39.7152
15	Polygon	3	4	agriculture	155.14	3	26	3	Builtup area	11.0207	Builtup area -agriculture	3.18144
16	Polygon	3	4	agriculture	155.14	4	111	1	Waterbody	14.6115	Waterbody -agriculture	0.051288
17	Polygon	4	5	barran land	123.662	1	3	2	Forest	271.02	Forest -barran land	6.65387
18	Polygon	4	5	barran land	123.662	2	4	4	Agriculture	151.814	Agriculture -barran land	41.7325
19	Polygon	4	5	barran land	123.662	3	26	3	Builtup area	11.0207	Builtup area -barran land	3.1854
20	Polygon	4	5	barran land	123.662	4	111	1	Waterbody	14.6115	Waterbody -barran land	0.472679

Graph of LULC Change 2016 & 2024

